

LeoEM

ISL routing emulation

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Stage 1

- Constellation Spatial Data
 - Using Starlink data
 - Satellite number: 1588
 - Cycles: 5731

Stage 2

- Route computation
 - Using ISL
 - Route
 - precomputed route from SD to Shanhei
 - compute route from Taipei to New York

Stage 3

- Emulation

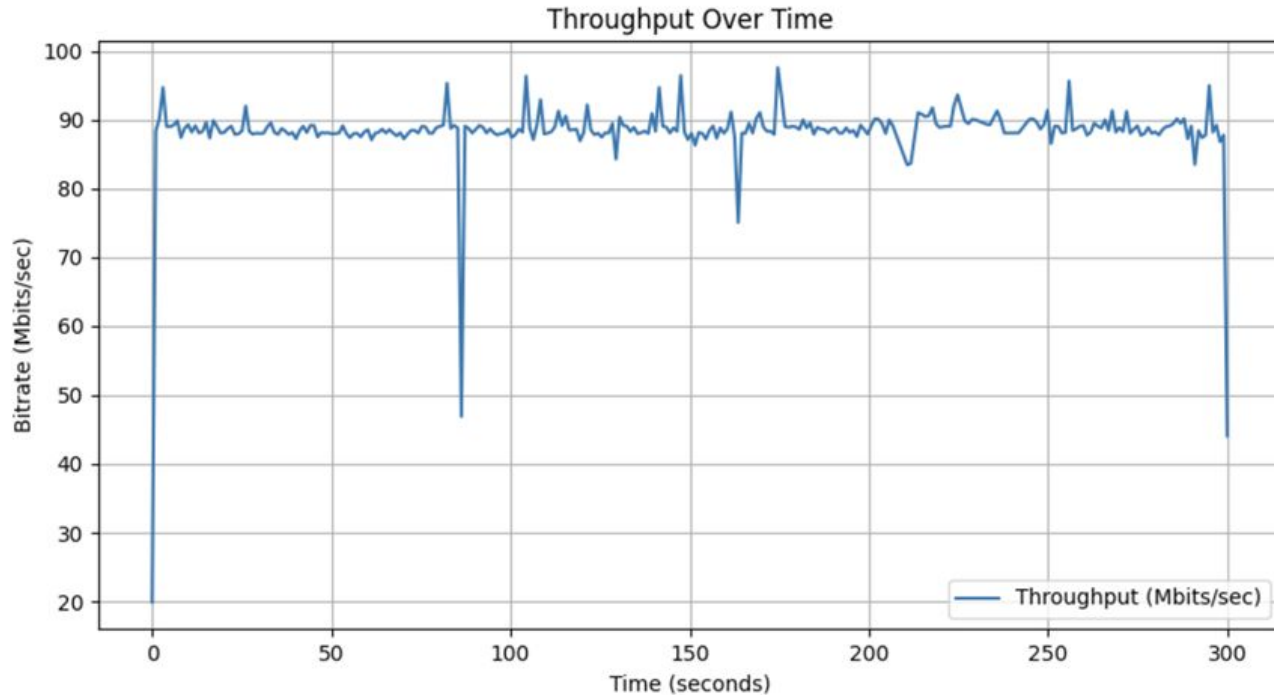
- Using precomputed route from SD to Shanhei
- Testing throughput and RTT from two endpoints
- Setting up SaTCP
- Testing with SaTCP

Emulation

- Setup
 - Ubuntu Server 24.04 on VM
 - Use Starlink_SD_Shanghai_15_ISL_path.log as route
 - Test throughput using iPerf3
 - h1(10.0.1.101) as client
 - h2(10.0.1.102) as server
 - Test RTT using tshark to capture iPerf traffic
 - Log handover time for analysis

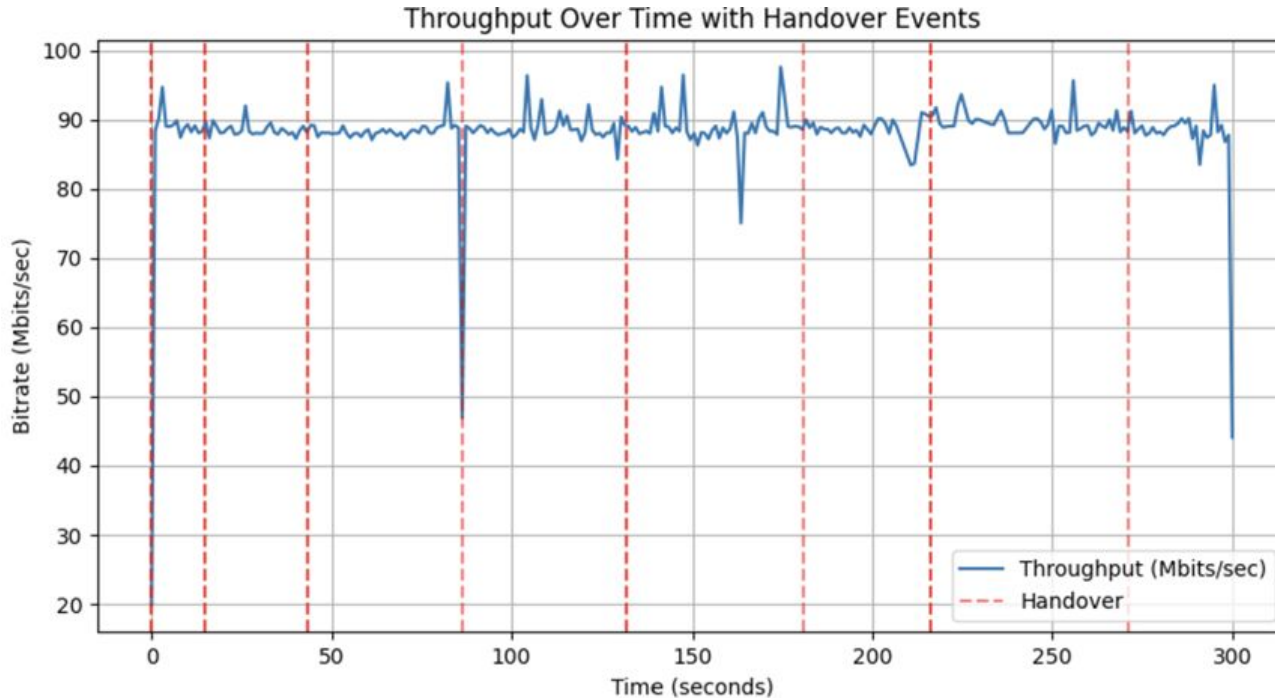
Emulation Result (TCP)

- Throughput result



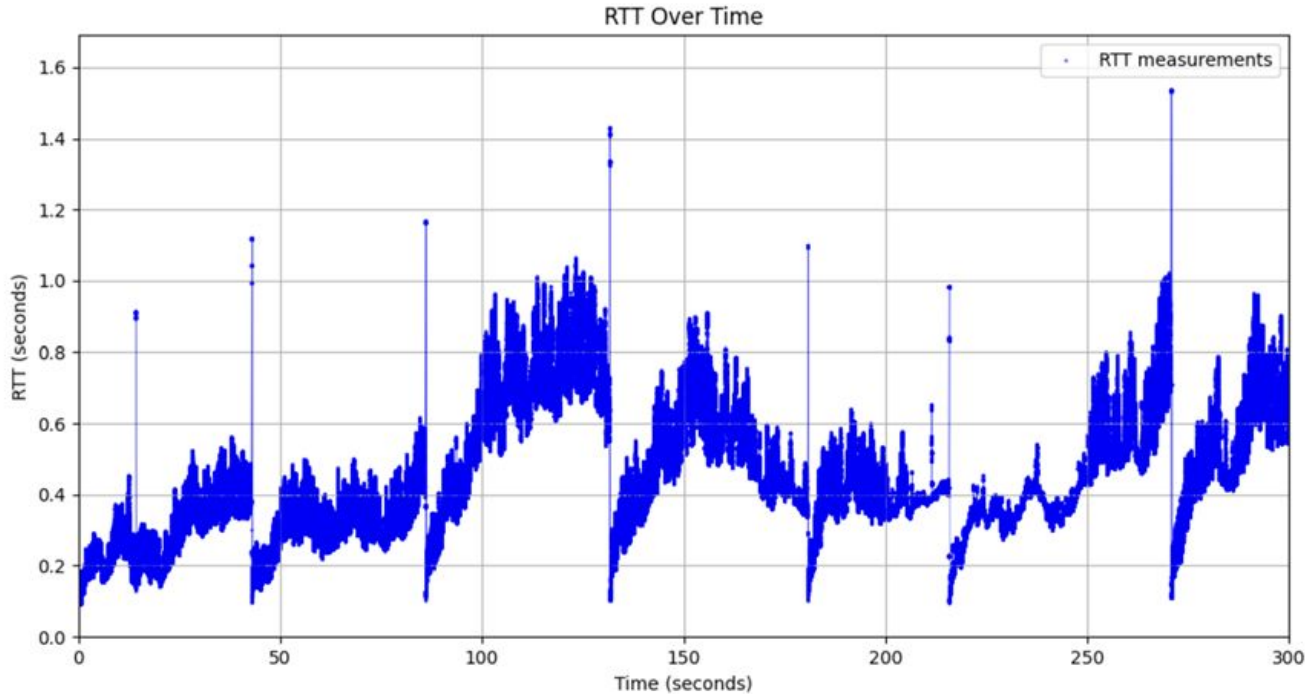
Emulation Result (TCP)

- Throughput result



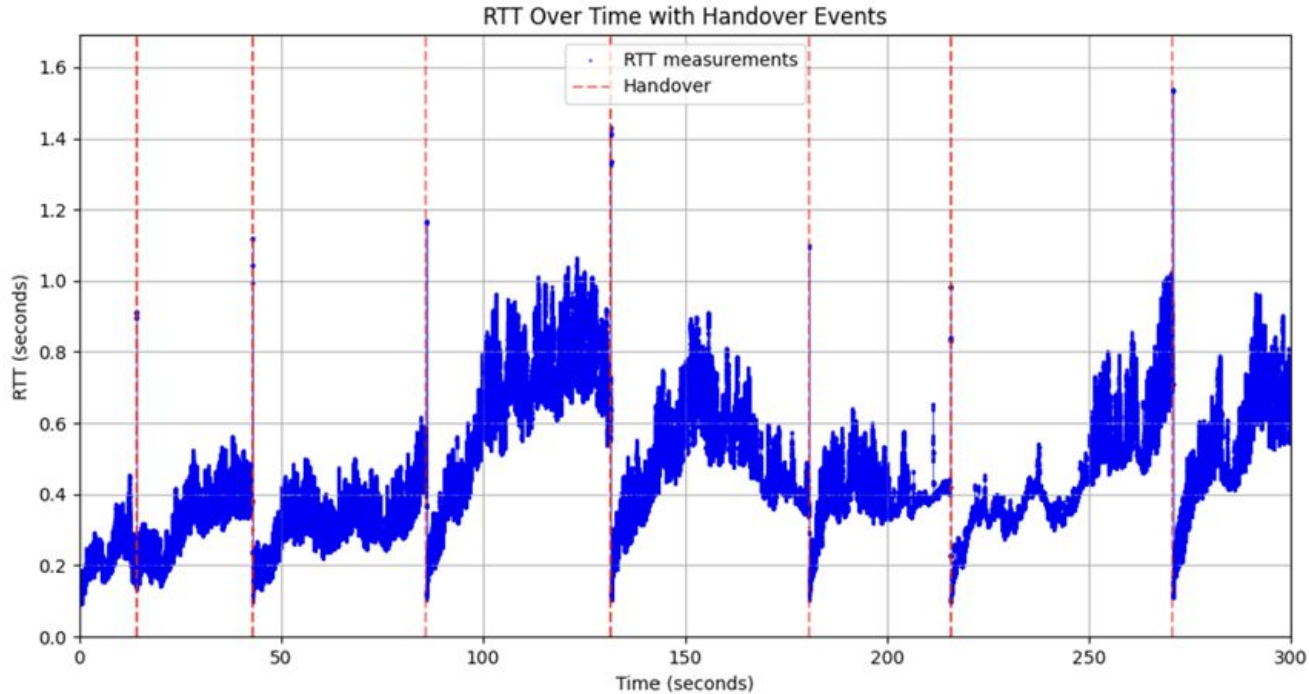
Emulation Result (TCP)

- RTT result



Emulation Result (TCP)

- RTT result

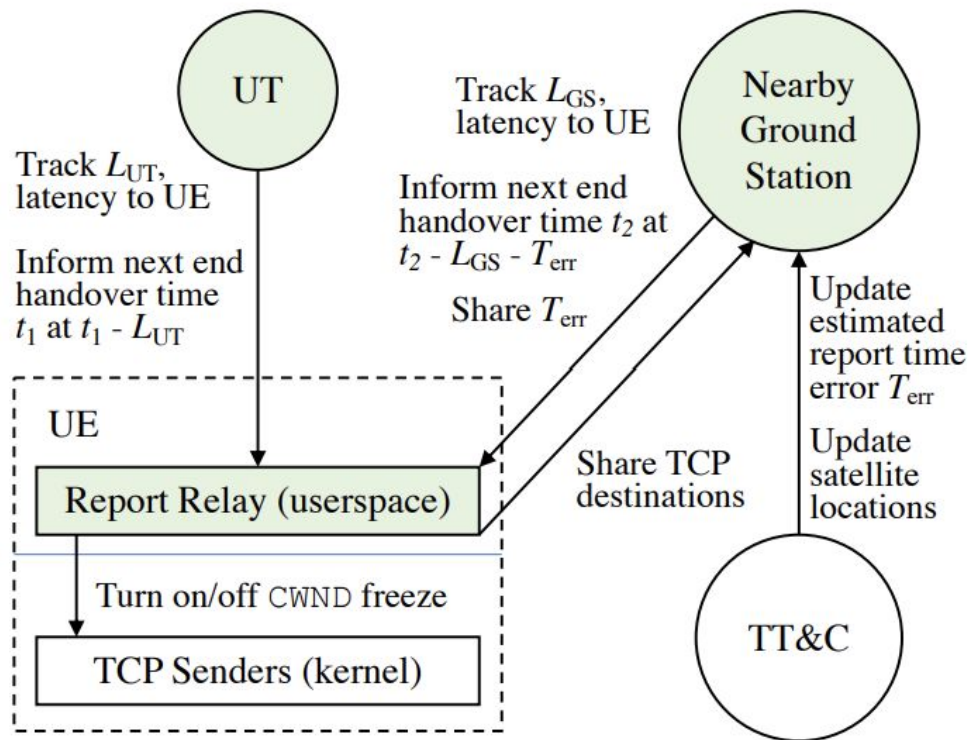


SaTCP

- Features of SaTCP [1]
 - Prediction of satellite location
 - Compute when a disruptive event will occur
 - Freeze congestion window during handover
 - Cross-Layer Integration
 - Collect real-time data from the link layer

SaTCP

- Workflow [1]

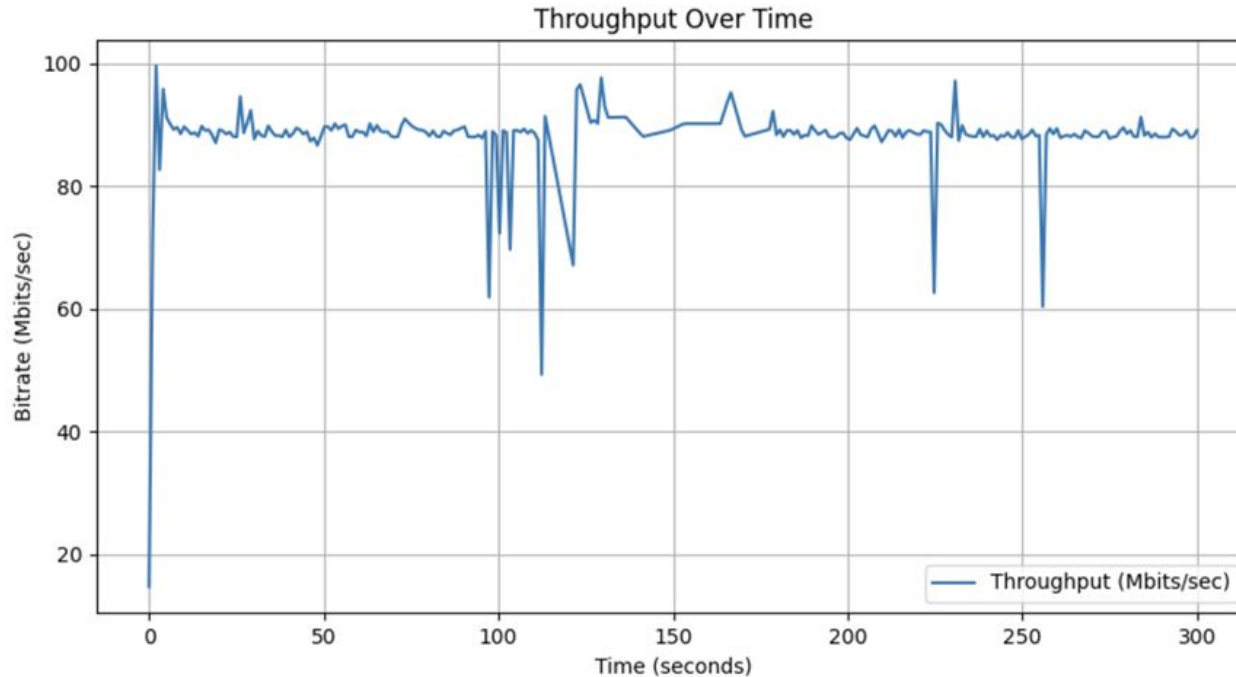


SaTCP

- Setup
 - Install Linux kernel source code 6.9.6 on VM
 - Follow the instructions of SaTCP Implementation in LeoEM
 - Run the emulation and tests

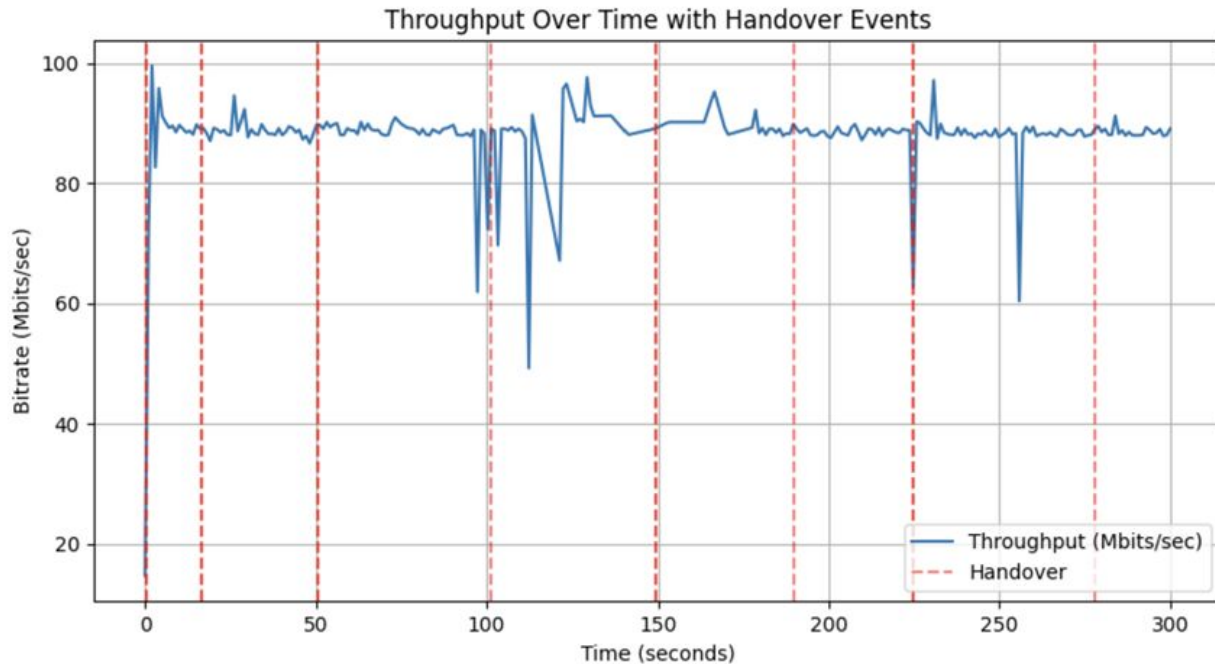
Emulation Result (SaTCP)

- Throughput result



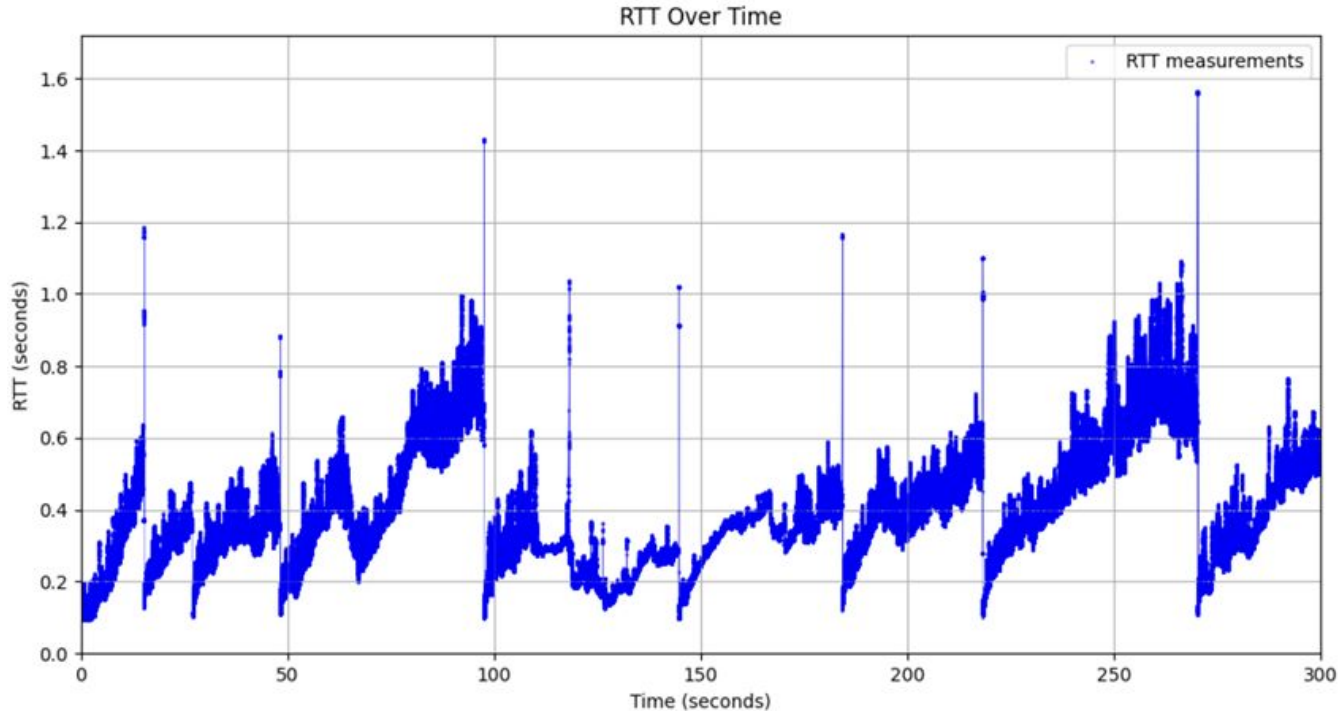
Emulation Result (SaTCP)

- Throughput result



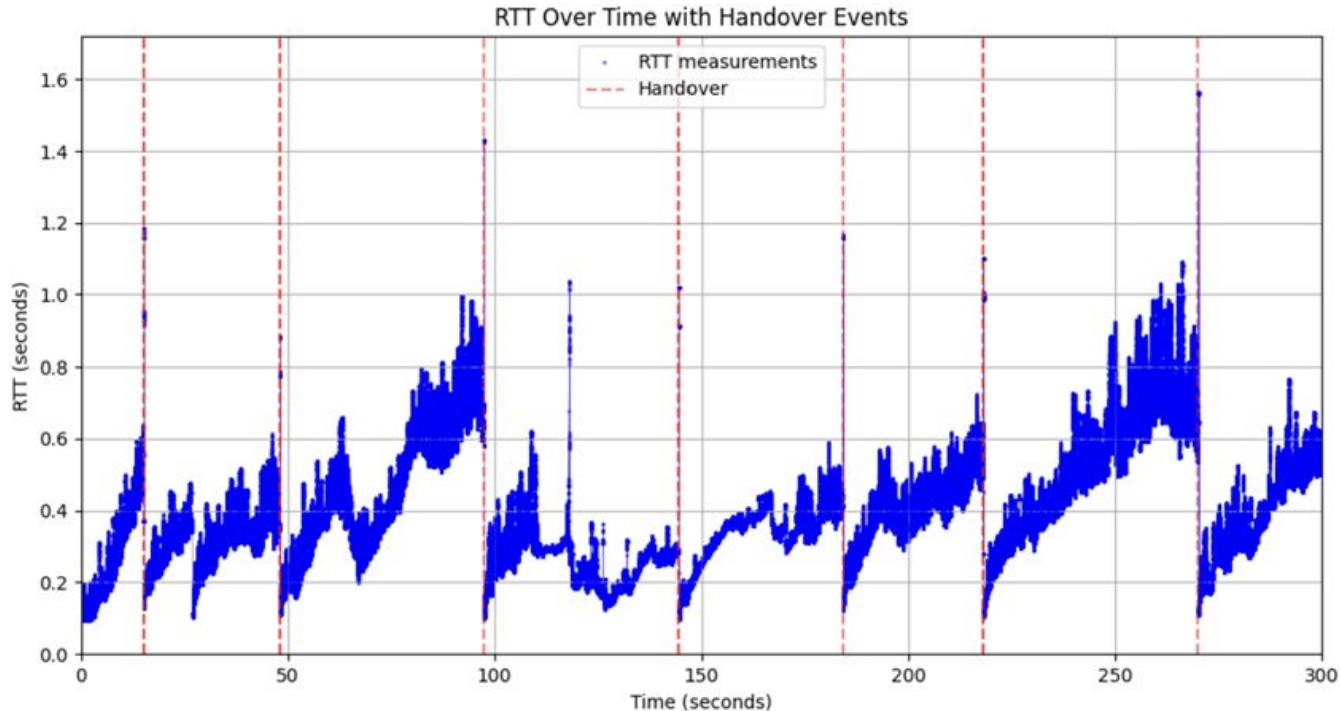
Emulation Result (SaTCP)

- RTT result



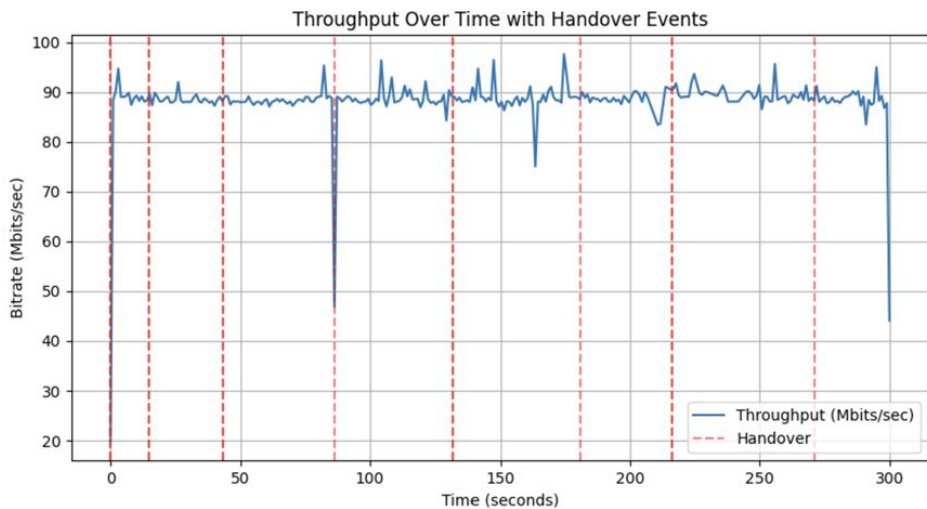
Emulation Result (SaTCP)

- RTT result

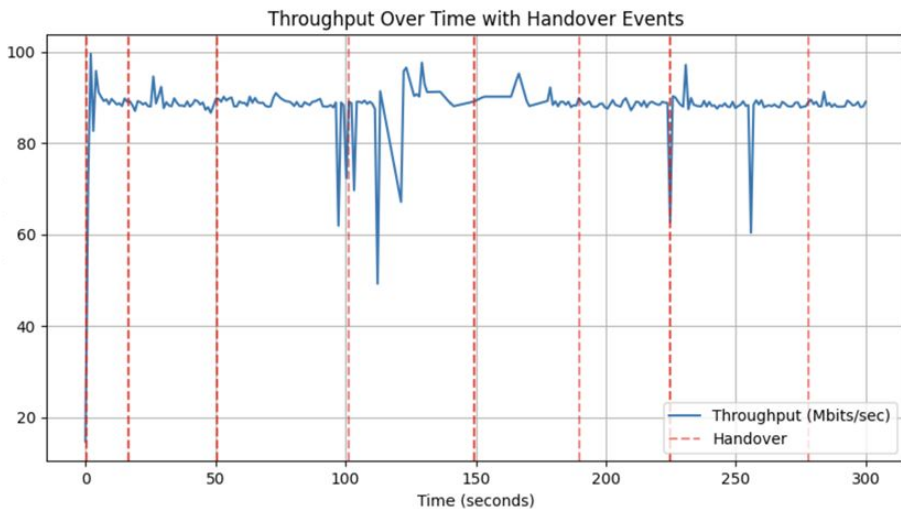


Comparison (Throughput)

TCP

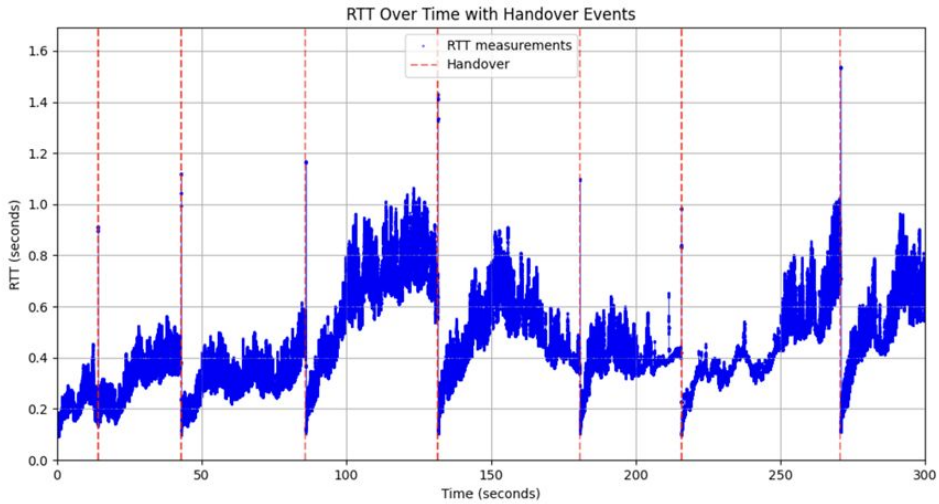


SaTCP

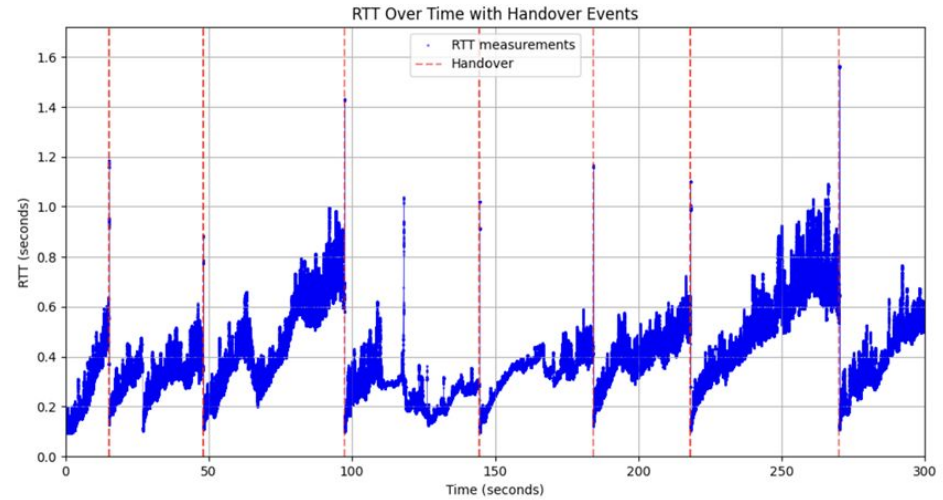


Comparison (RTT)

TCP



SaTCP



Comparison

	TCP	SaTCP
Average Throughput (Mbits/sec)	89.84	88.05
MinimumThroughput (Mbits/sec)	47.94	14.70
Average RTT (sec)	0.44	0.38
Maximum RTT (sec)	1.54	1.56

The End